Taadeen Automation — Code Guide

What every file does, why it matters, and the locators it uses

67
files / file groups

127
locators documented

Playwright + TypeScript
OutSystems Reactive app

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Architecture

The code is layered so each change has one home: a new screen is a page object, a new widget a component, a new business rule a flow. Specs stay short and describe *what* is tested.

Layer	Responsibility
`tests/`	Specs — what is tested. Short, readable, configure the run (environment, users, licences).
`src/flows/`	Business flows across screens and users (complete a draft, drive the approval workflow).
`src/pages/`	Page objects — one class per screen, exposing actions and locators. No business assertions.
`src/components/`	Reusable OutSystems widgets (wizard, popup, multi-select, toast, row menu) used by many pages.
`src/utils/`	Helpers with no screen of their own: server-call waits, field helpers, KML and data generators, logging.
`src/fixtures/`	The Playwright `test` object every spec imports: logged-in investor / employee per test.
`src/auth/`	Login and session reuse: one browser context per user, saved sessions, login lock.
`src/config/`	Environments, portals, users, run options and production safety — all read from .env.
`src/data/`	Test data defaults (accepted by TEST) with JSON overrides per environment.
`scripts/`	Command-line helpers: env check, clear sessions, generate KML, record a scenario.

How a test runs (new Exploration License)

1. Spec sets RUN (env, licences, users) and calls test.use({ investorUser, employeeUser }).
2. fixtures/test.ts → SessionManager logs each user in (or reuses a saved session) in its own browser context.
3. Investor: RequestsPage → "طلب جديد" → ExplorationLicensePage wizard; each step page object fills its screen.
4. Step 1 save returns the request number (ActionSave_licenseInfo → LicenseReqId).
5. Step 2: KML generated → uploaded → GIS map iframe checks it → "تأكيد" saves the location (retries on overlap).
6. Steps 3–8 filled by ExplorationLicenseFlow, then "إرسال" submits.
7. ApprovalWorkflow loops: employee task (TaskManagementPage → TaskResponsePage) or investor payment (ToDoListPage → PaymentPage → PaymentNotificationPage) until status "مكتمل".

Locator strategy

- **Accessible roles and visible text first** — `getByRole('button', { name: 'بحث' })`, `getByPlaceholder`, `getByLabel`. They match what the user sees and survive layout changes.
- **OutSystems block names for sections** — `[data-block="WorkPlanDescription"]`: stable, language-independent and unique per section.
- **Column titles for table cells** — `td[data-header="رقم الطلب"]`: survives column re-ordering.
- **Widget-name suffixes for footer buttons** — `button[id$="-Next"]`: the suffix is the widget name; the block prefix (e.g. `b6-`) changes.
- **Visible-only filters** — `.filter({ visible: true })` where a screen keeps hidden duplicates (second table, other tab).
- **Exact text, never position** — decisions and options are chosen by their label; unknown options stop the run instead of guessing.

- **Server responses as signals** — waiting for a named OutSystems action (e.g. `ActionSave_licenseInfo`) instead of sleeps or networkidle.

Tests (specs)

`tests/exploration-license/new-exploration-license.spec.ts`

PURPOSE Creates a NEW Exploration License from scratch and completes it end to end: Requests → "طلب جديد" → steps 1–8 → submit → all employee tasks, both payments and approvals until "مكتمل".

WHY IT MATTERS The main end-to-end test. Its RUN block is the control panel of a run: env (test/uat), number of licences created in parallel, and which investor / employee log in. AUTHORISED_ENVS blocks environments where creating real requests is not allowed; prod is always refused. LICENSE_DETAILS sets step 1 values (category, duration, minerals, mining complex) and GIS_MAP_TIMEOUT_MS how long the map may take.

Locator	What it finds
`waitForResponse(url includes '/ActionSave_licenseInfo')`	Step 1 save — its JSON returns LicenseReqId, the new request number (safe in parallel runs).

`tests/exploration-license/exploration-license-issuance.spec.ts`

PURPOSE Continues an EXISTING request (REQUEST_NUMBER) from its current state: completes the draft if needed, then runs the same approval workflow to the end.

WHY IT MATTERS Resumable — re-running picks up wherever the request is. Useful for finishing requests left half-way. USERS chooses the investor (must own the request) and employee. Guarded to TEST only.

`playwright.config.ts`

PURPOSE Central Playwright configuration: test folder, timeouts per environment, workers, reporters (list, HTML, JUnit, Allure), browser project (chromium), trace/video/screenshot policy.

WHY IT MATTERS PROD guard #1: on prod only @prod-safe tests run. WORKERS / HEADLESS / RECORD come from .env. No URLs or credentials here, so `playwright test --list` works on a fresh clone.

Flows (business processes)

`src/flows/approval-workflow.flow.ts`

PURPOSE Drives one request from submission to the end of its approval workflow. Each round it looks at what is pending — an open employee task, otherwise an investor To-Do task — and handles exactly that.

WHY IT MATTERS State-driven, not a fixed script, so a re-run resumes where the request is. Decisions are chosen by option text, never by position (e.g. أرض متاحة, قبول التوصية, ممتثل, التوصية بالقبول, قبول الطلب, تحققت المتطلبات). Unknown option sets stop with UnsupportedTaskError. Payments: acknowledge → capture Bill ID → payment notification → verify the "الدفع" task left the To-Do list. Paid bills are stored in .workflow-state/<request>.json so a bill is never paid twice. Ends on "مكتمل" (or EXPECTED_FINAL_STATUS); dead states مرفوض / ملغي stop the run.

`src/flows/exploration-license.flow.ts`

PURPOSE Completes the Exploration License wizard: open a request, pass steps 1–2 unchanged (existing drafts), fill the current step, Next, Submit, read the status in the list.

WHY IT MATTERS Dispatches each step by the wizard's own step name, so a renamed or new step fails loudly instead of being filled with guessed data.

`src/flows/request.flow.ts`

PURPOSE Contract (interface) for a generic investor-submits / employee-acts flow.

WHY IT MATTERS Not implemented on purpose — placeholder that throws NotImplementedError until those screens are explored.

Investor pages

src/pages/investor/InvestorPortal.ts

PURPOSE Entry point to all investor page objects for one logged-in session: investor.login, .home, .requests, .explorationLicense.

WHY IT MATTERS Tests never call `new XPage(page)`; new investor screens are added here as getters.

src/pages/investor/InvestorLoginPage.ts

PURPOSE Investor login screen (<ENV>_INVESTOR_URL, /MiningPortal/Login).

WHY IT MATTERS Only declares the selectors; the login steps live in LoginPage. Verified on UAT.

Locator	What it finds
`#Input_UsernameVal2`	Username (email) field.
`#Input_PasswordVal2`	Password field.
`button[type="submit"]`	"تسجيل الدخول" button.

src/pages/investor/InvestorHomePage.ts

PURPOSE Landing page after investor login.

WHY IT MATTERS Proves "authenticated" generically (login form gone) until a dashboard landmark is chosen.

Locator	What it finds
`input[type="password"] (count 0)`	No password field left = logged in.

src/pages/investor/RequestsPage.ts

PURPOSE Investor "الطلبات" list (/MiningPortal/Requests): open the list, search a request, read its status, start "طلب جديد", run a row action (عرض / حذف / تعديل).

WHY IT MATTERS The screen holds two tables and two search boxes (one hidden tab), so every locator is limited to visible elements.

Locator	What it finds
`getByRole('link', { name: /الطلبات\s*$/ }).first()`	Side-menu entry "الطلبات".
`getByPlaceholder('اكتب رقم الطلب').filter({ visible: true })`	Search box (visible one).
`getByRole('button', { name: 'بحث', exact: true }).filter({ visible: true })`	Search button → server action DataActionGetLRData.
`locator('table').filter({ visible: true }).first()`	The visible requests table.
`tbody tr has td[data-header="رقم الطلب"] = <n>`	Row of one request (cells carry data-header = column title).
`td[data-header="الحالة"] / "الإجراء"`	Status cell / actions cell of a row.
`getByRole('button', { name: /طلب جديد/ })`	"طلب جديد" (new request) button.
`getByRole('dialog').last() → getByRole('combobox')`	Service type select in the "إنشاء طلب جديد" popup.
`dialog.getByRole('button', { name: 'التالي', exact: true })`	Popup Next → opens the new wizard.
`[aria-describedby] → [id="<balloon>"], .esnad-dots`	Row "⋯" menu: icon and its tooltip balloon.
`menu.getByRole('link', { name: action, exact: true })`	A row action link (عرض, حذف, تعديل).

src/pages/investor/ToDoListPage.ts

PURPOSE Investor "قائمة المهام" (To-Do list): find the tasks of a request (e.g. "الدفع") and open one ("تعيين").

WHY IT MATTERS Where the investor's payment tasks appear; the workflow uses it to detect and verify payments.

Locator	What it finds
`getByRole('link', { name: /قائمة المهام\s*$/ })`	Side-menu entry.
`getByRole('spinbutton', { name: 'اكتب رقم الطلب' })`	Search box (numeric).
`getByRole('button', { name: 'بحث', exact: true })`	Search button.
`tbody tr has td[data-header="الرقم التعريفي للطلب"]`	Rows of one request.
`td[data-header="نوع المهمة"]`	Task type cell (e.g. "الدفع").
`row td:last-child → RowActionMenu`	Row menu ("تعيين" opens the payment page).

src/pages/investor/PaymentPage.ts

PURPOSE Payment page (SharedServices_Payments/DoPayment): read the bill (SADAD reference, amount), tick the acknowledgement and confirm.

WHY IT MATTERS Captures the Bill ID that the payment notification then marks as paid.

Locator	What it finds
`waitForURL(/DoPayment\?/)`	Payment page opened.
`getByText(/الرقم المرجعي لنظام سداد/)`	SADAD reference label (bill number next to it).
`locator('body').innerText()`	Page text, parsed for bill id and amount.
`getByRole('checkbox').filter({ visible: true }).first()`	Acknowledgement checkbox.
`getByRole('button', { name: 'تأكيد', exact: true })`	Confirm button.

Exploration License wizard (investor)

src/pages/investor/exploration-license/ExplorationLicensePage.ts

PURPOSE The wizard screen ("طلب رخصة إستكشاف جديدة") and its 8 step page objects: requestInfo, location, workPlan, financialCapability, socialImpact, environmentalImpact, attachments, terms.

WHY IT MATTERS Different OutSystems module from the portal, so it is reached through the Requests list, never by building the URL. Cold load ~13 s, so first render has a 60 s budget.

Locator	What it finds
`waitForURL(/Exploration_License\?/, commit)`	New wizard opened.
`waitForURL(…In_RequestId=<n>…)`	Existing request's wizard opened.

src/pages/investor/exploration-license/WizardStep.ts

PURPOSE Base class of every wizard step: fill() wraps fillStep() and returns a StepReport of what was filled vs. kept.

WHY IT MATTERS Sections are found by their OutSystems block name (data-block) — stable, language-independent and unique. Never overwrites existing data, so re-runs are safe.

Locator	What it finds
`[data-block="<SectionName>"]`	A step section, e.g. WorkPlanDescription, Drilling, RehabilitationPlan.
`section tbody tr`	Rows already added to a section table ("kept" if any).
`section.getByRole('button', { name: /إضافة/ })`	Section "إضافة" button → AddRowDialog.

src/pages/investor/exploration-license/RequestInfoStep.ts

PURPOSE Step 1 "معلومات الطلب" on a new request: declarations, licence category, duration, minerals, mining complex, one geologist.

WHY IT MATTERS Values come from LICENSE_DETAILS in the spec. The minerals list loads from the server after the category, so it waits for server idle. Geologist popup needs 3 attachments (uploadAll).

Locator	What it finds
`[role=tab] has getByRole('button', { name: <title> })`	Accordion section by its title (التزامات وإفصاحات مقدم الطلب, القدرة الفنية, مدة الرخصة والمعادن المطلوبة).
`getByRole('button', { name: 'تطبيق على الكل', exact: true })`	Tick all declarations.
`getByRole('checkbox', { checked: false }) (count 0)`	Every declaration ticked.
`getByRole('combobox', { name: 'Select an option' })`	فئة الرخصة (single Virtual-Select).
`licence.locator('select').nth(0) / .nth(1)`	مدة الرخصة / المجمع التعديني (native selects).
`getByRole('combobox', { name: 'Select one or more options' })`	المعادن (multi-select with search).
`getByRole('button', { name: /إضافة جديدة/ })`	Add geologist → popup "إضافة خبير جيولوجي جديد".
`technical tbody tr`	Geologist row added.

src/pages/investor/exploration-license/LocationStep.ts

PURPOSE Step 2 "موقع الطلب": generate a clockwise KML, upload it, open the GIS map, wait for its verdict, click "تأكيد", retry with a new KML on errors.

WHY IT MATTERS The most fragile step. Traps: "تأكيد" starts enabled before validation finishes (must stay enabled 4 s); "تأكيد" saves via 3 slow applyEdits calls, so the success message ("نجاح") must be awaited (up to mapTimeoutMs); errors "تقاطع المواقع" / counter-clockwise trigger a retry; the map may sit below the viewport (close button is clicked via dispatchEvent). Every attempt is attached as location-attempts.json.

Locator	What it finds
`locator('label').filter({ hasText: 'تحميل ملف' })`	Choose "upload file" (label covers the radio).
`getByRole('radio', { name: 'تحميل ملف' })`	Radio state check.
`input[type=file][accept=".KML"]`	KML upload input (max 1 MB).
`waitForResponse(url includes 'ActionBinaryDataToText')`	KML parsed into coordinate rows.
`getByText('انظر على الخريطة', { exact: true })`	"See on map" → loads the map iframe.
`page.frames() url matches /MapComponent\//`	GIS iframe ugisw.../MapComponent/Outside|Inside.html.
`frame #btnSubmit`	"تأكيد" (confirm) inside the map.
`frame getByRole('heading', { level: 2 })`	Map error messages (e.g. تقاطع المواقع).
`frame getByRole('button', { name: 'إغلاق' })`	Close a map error popup.

src/pages/investor/exploration-license/WorkPlanStep.ts

PURPOSE Step 3 "خطة العمل و الانفاق": technical justification (5 attachments), work-plan description, field survey and land ownership radios, study / survey / lab / expenses popups, drilling, rehabilitation cost per year, development timeline, declarations.

WHY IT MATTERS Largest step; values from exploration-license.data.ts. Mandatory rules (as reported by the app) are listed in the file header.

Locator	What it finds
`[data-block="TechnicalJustification"]`	المبررات الفنية section (popup with 5 attachments).
`[data-block="WorkPlanDescription"] [role=combobox] [aria-haspopup=listbox]`	طبيعة عمليات الكشف multi-select.
`WorkPlanDescription getByRole('radiogroup')`	نتائج أعمال المسح الميداني radios.
`[data-block="LandOwnership"]`	ملكية الأرض radios.
`[data-block="StudyPlan" | "TechnicalStudy" | "SurveyPlan_GeochemicalAnalysis" | "SurveyPlan_Geophysical" | "PhysicalLaboratoryPlan" | "OtherExpensesPlan"]`	Plan sections filled through AddRowDialog.
`[data-block="Drilling"] has .font-size-h6 = "الحفر الماسي" / "حفر الدوران العكسي"`	Drilling sub-sections by heading.
`[data-block="RehabilitationPlan"] tbody input`	Rehabilitation cost per year.
`[data-block="DetailsOfTheDevelopment"] tbody tr → select, input[type=text]`	Timeline rows: start year/month, duration.
`[data-block="ApplicantsDeclaration"] checkbox`	Declarations.

src/pages/investor/exploration-license/FinancialCapabilityStep.ts

PURPOSE Step 4 "إثبات القدرة المالية": proof type, available balance, attachment.

WHY IT MATTERS Checkboxes have no accessible names, so they are addressed by screen order (الشهادة البنكية, القوائم المالية, خطاب تسهيلات, كشف الحساب البنكي).

Locator	What it finds
`section.getByRole('checkbox') (4, by index)`	Proof type.
`section.getByRole('spinbutton')`	الرصيد المتاح (available balance).
`[data-block$="UploadAttachment"] (visible)`	Attachment slot.

src/pages/investor/exploration-license/SocialImpactStep.ts

PURPOSE Step 5 "خطة الأثر الاجتماعي": community employee details, local communities, grievance, stakeholders, impacts, programmes, local employment and purchase plans.

WHY IT MATTERS Field formats are enforced by the app (ID 10 digits starting 1/2, mobile 9 digits starting 5, valid email) — values come from data-generator.ts.

Locator	What it finds
`[data-block="EmployeeAppointedtoManageCommunityPerformance"]`	Community employee fields.
`[data-block="LocalCommunitiesandEngagement" | "SocialImpactAssessmentPlan" | "EmploymentofLocalCommunityMembersPlan" | "PurchaseplansfromtheLocalCommunity"]`	Plan sections.
`[role=combobox][aria-haspopup=listbox] .nth(i)`	Multi-selects by position.

src/pages/investor/exploration-license/EnvironmentalImpactStep.ts

PURPOSE Step 6 "خطة الأثر البيئي": mining impacts popup (per domain: impacts, causing activities, mitigation, monitoring) and applicant obligations.

WHY IT MATTERS Labels repeat per domain, so controls are addressed by domain position.

Locator	What it finds
`[data-block="ImpactsCausedbyMiningActivities"] .table-custom-dga > .gridContainer6:not(.table-header-custom)`	Impact rows already added.
`dialog [role=combobox][aria-haspopup=listbox]`	3 multi-selects per domain.
`dialog locator('select').nth(domain)`	Monitoring frequency.
`[data-block="ApplicantObligations"] checkbox`	Obligations.

src/pages/investor/exploration-license/FinalSteps.ts

PURPOSE Step 7 "المرفقات" (optional attachments, passed empty) and step 8 "الشروط والأحكام" (accept terms).

WHY IT MATTERS Last step before "إرسال" (submit).

Locator	What it finds
`section.getByRole('checkbox')`	Terms acceptance checkbox(es).

Employee pages

src/pages/employee/EmployeePortal.ts

PURPOSE Entry point to employee page objects for one session: employee.login, .home, task pages.

WHY IT MATTERS Same pattern as InvestorPortal.

src/pages/employee/EmployeeLoginPage.ts

PURPOSE Employee LDAP login screen (/MiningPortal/Login_LDAP).

WHY IT MATTERS IDs differ from the investor screen (...Val vs ...Val2) — one reason each portal has its own login page.

Locator	What it finds
`#Input_UsernameVal`	LDAP username.
`#Input_PasswordVal`	Password.
`button[type="submit"]`	"تسجيل الدخول".

src/pages/employee/EmployeeHomePage.ts

PURPOSE Landing page after employee login.

WHY IT MATTERS Generic "logged in" check.

Locator	What it finds
`input[type="password"] (count 0)`	Login form gone.

src/pages/employee/TaskManagementPage.ts

PURPOSE Employee "إدارة المهام" (/SharedServices_TaskManagement/TaskManagement): search a request, list its open tasks (جديد / قيد التنفيذ), assign a task to self, open "الرد على الطلب".

WHY IT MATTERS On UAT the menu entry is hidden in a collapsed group, so the page is opened by URL when the entry is not visible. All tab tables are in the DOM; the Licenses table is the first.

Locator	What it finds
`getByText('إدارة المهام', { exact: true }).filter({ visible: true })`	Side-menu entry (TEST); else goto the URL (UAT).
`locator('table').first()`	Licenses tasks table.
`getByText(/^البحث\(\d+\)/)`	Search panel toggle.
`getByPlaceholder('رقم الطلب', { exact: true })`	Search box.
`getByRole('button', { name: 'البحث', exact: true })`	Search → server action DataActionGetData.
`td[data-header="رقم الطلب" | "نوع المهمة" | "الحالة" | "مسند الى" | "الإجراء"]`	Task row cells.
`dialog label 'لي' → getByRole('radio', { name: 'لي' })`	Assign to me.
`dialog getByRole('button', { name: 'حفظ', exact: true })`	Save → ActionAssign_Self; status becomes قيد التنفيذ.

src/pages/employee/TaskResponsePage.ts

PURPOSE Task page opened by "الرد على الطلب" (new tab, /Licenses_Explorations/Request_Response): choose the decision, tick checklists, untick government approvals, fill "الوصف", submit "إرسال الرد".

WHY IT MATTERS Decision radios are chosen by label text; a successful submit returns the tab to Task Management, a refused one shows inline "هذا الحقل مطلوب".

Locator	What it finds
`getByRole('tab', { name: 'الرد على الطلب' })`	Respond tab.
`getByRole('radiogroup') has getByRole('radio', { name: <decision> })`	Decision group (e.g. أرض متاحة, قبول التوصية).
`group label = <decision>`	Click the label (it covers the input).
`radiogroup has radio 'تحققت المتطلبات'`	Multi-part checklist: every item met.
`a[id$="-Next2"]`	Next part of a multi-part assessment.
`getByRole('combobox', { name: 'Select one or more options' })`	Government entities (if configured).
`respondPanel getByRole('checkbox')`	e.g. "الأرض بحاجة إلى الموافقات الحكومية" (unticked).
`getByRole('textbox', { name: 'اكتب هنا' })`	"الوصف" description.
`getByRole('button', { name: 'إرسال الرد', exact: true })`	Submit the response.
`waitForURL(path ends '/TaskManagement')`	Submit accepted.

Shared pages and test utilities

src/pages/BasePage.ts

PURPOSE Base class of every page object: knows its portal (URLs from config), OutSystems server-call helpers, shared feedback toast.

WHY IT MATTERS Keeps waits consistent across pages; no business assertions here.

src/pages/LoginPage.ts

PURPOSE Generic username/password login: fill, submit, wait until the URL leaves the login path, classify the outcome (success / wrong credentials / …).

WHY IT MATTERS Portal login pages only supply selectors; LoginFailedError explains failures.

Locator	What it finds
`selectors.username / password / submit`	Supplied by InvestorLoginPage / EmployeeLoginPage.
`waitForURL(path ≠ login path)`	Login succeeded.

src/pages/login-paths.ts

PURPOSE Login paths of every configured portal; isAnyLoginPage(url).

WHY IT MATTERS An expired session may redirect to the other portal's login, so any login path means "not authenticated".

src/pages/test-utils/PaymentNotificationPage.ts

PURPOSE Test-only payment utility (<host>/TadeenTest/CallPaymentNotification): marks a bill as paid by sending the page's own template with the bill number.

WHY IT MATTERS Allowed environments from PAYMENT_UTILITY_ENVS (default test,uat; never prod). Provider id from <ENV>_PAYMENT_SERVICE_PROVIDER_ID, else the template value. Reads the server answer: anything but S000 fails immediately. Trap: the page note "UAT: 004" is wrong — UAT accepts its template value 002.

Locator	What it finds
`getByRole('button', { name: 'Call Notification' })`	Send button.
`locator('body').innerText()`	Template JSON and printed response.
`locator('textarea').first()`	Request body field.
`waitForResponse(url includes '/ActionBill_Req')`	Server answer: data.Response.Status (S000 = success).

src/pages/index.ts

PURPOSE Barrel file re-exporting the page objects.

WHY IT MATTERS Shorter imports.

Components (reusable widgets)

src/components/Wizard.ts

PURPOSE OutSystems "Wizard": the step list plus the Previous / Next / Submit footer. next() reports whether the step moved and the validation errors if not.

WHY IT MATTERS Next runs client-side validation first (error toast, no server call); success saves the step server-side. Clicking a step does not jump forward.

Locator	What it finds
`getByRole('tablist') has .wizard-wrapper-item`	The step list.
`[role=tab].wizard-wrapper-item:not(.hideContainer)`	Visible steps (label in aria-label; .active current, .past done).
`button[id$="-Next"]`	Next (id suffix is stable, prefix like "b6-" is not).
`button[id$="-Submit"]`	Submit ("إرسال").
`feedback.of('error') (visible)`	Validation error toast.

src/components/AddRowDialog.ts

PURPOSE "إضافة" popup used by the wizard's table sections: fill labelled fields, upload every attachment (uploadAll), save, report validation errors.

WHY IT MATTERS Two layouts (line-based with an inner "إضافة", and single-form). uploadAll counts the file inputs once — the popup can briefly show none while re-rendering. A popup that stays open is reported as an error.

Locator	What it finds
`getByRole('dialog').last()`	The open popup.
`getByRole('button', { name: 'إضافة', exact: true })`	Add-line / save buttons (last one saves).
`locator('input[type=file]')`	Attachment inputs (each upload removes one).
`tbody tr`	Lines added inside the popup.
`[class*=close]`	Close (X).

src/components/MultiSelect.ts

PURPOSE OutSystems "Dropdown Tags" (Virtual-Select): open, list options, select by exact label (using search for the virtualised list), close.

WHY IT MATTERS Options live in a separate element (aria-controls); Escape does not close it. Single-selects close themselves on click, so clearing the search is optional.

Locator	What it finds
`[role=combobox][aria-haspopup=listbox]`	The widget (passed in by the page).
`#<aria-controls id>`	Its dropdown box.
`.vscomp-toggle-button`	Open / close.
`.vscomp-option (exact text)`	An option.
`input.vscomp-search-input`	Search inside the list.

src/components/FeedbackMessage.ts

PURPOSE OutSystems toast messages after server actions: count() before an action, textsSince() after it.

WHY IT MATTERS Toasts stack and stay visible, so "an error is visible" is not proof of a NEW error.

Locator	What it finds
`.feedback-message-<success|error|warning|info>`	Toast of a type.
`.feedback-message-text (visible)`	Toast texts.

src/components/RowActionMenu.ts

PURPOSE The "⋯" menu of a table row: list and click its actions.

WHY IT MATTERS Shared by Requests, Task Management and To-Do list. Disabled entries are plain text, not links.

Locator	What it finds
`actionCell [aria-describedby]`	Menu trigger.
`[id="<balloon id>"]`	The tooltip balloon.
`balloon.getByRole('link', { name, exact: true })`	An action.

src/components/Dropdown.ts

PURPOSE Generic dropdown: native <select> or Virtual-Select search dropdown.

WHY IT MATTERS Generic helper; the wizard mostly uses MultiSelect and native selects directly.

Locator	What it finds
`select / option`	Native dropdown.
`.vscomp-toggle-button, .vscomp-dropbox-container:visible, input.vscomp-search-input, .vscomp-option`	Virtual-Select dropdown.

src/components/DataTable.ts

PURPOSE Generic table helper: rows, row by text.

WHY IT MATTERS Text-based lookups survive column re-ordering and dynamic ids.

Locator	What it finds
`tbody tr`	Rows.

src/components/FileUpload.ts

PURPOSE Upload into a (hidden) <input type="file">.

WHY IT MATTERS setInputFiles works on hidden inputs — no clicking styled buttons.

Locator	What it finds
`input[type="file"]`	File input.

src/components/BaseComponent.ts / index.ts

PURPOSE Base class (component scoped to a root locator) and barrel export.

WHY IT MATTERS Page objects compose components; tests never touch raw selectors.

Fixtures and authentication

src/fixtures/test.ts

PURPOSE The `test` / `expect` every spec imports. Fixtures: investor, employee (logged in), sessions (any other user), envConfig; options investorUser / employeeUser.

WHY IT MATTERS Lazy — a test only logs in the users it asks for. A spec picks its users with test.use({ investorUser, employeeUser }).

src/auth/SessionManager.ts

PURPOSE Creates one isolated browser context per user, reuses a saved session when still valid, otherwise logs in; records video per context.

WHY IT MATTERS Investor and employee never share cookies. Only one worker logs a user in at a time (login lock) — two simultaneous logins of the same user log each other out.

src/auth/AuthStateStore.ts

PURPOSE Saves sessions in .auth/<env>/<user>.json (+ meta) for AUTH_SESSION_TTL_MINUTES; cross-process login lock.

WHY IT MATTERS Faster runs, fewer LDAP logins. Atomic writes so parallel workers never read half a file. Git-ignored (contains cookies).

src/auth/strategies/*

PURPOSE LoginStrategy interface, FormLoginStrategy (username/password via the portal's LoginPage) and the registry strategyForPortal().

WHY IT MATTERS A new login mechanism (SSO, Nafath, OTP) is a new strategy class — no test changes.

Configuration

src/config/env.ts

PURPOSE Loads .env once; readVar, readBool, readNumber, requireVars (one clear message listing every missing variable).

WHY IT MATTERS One .env with environment-prefixed keys; real process variables (CI secrets) always win.

src/config/environments.ts

PURPOSE Environment registry (dev, sit, test, uat, prod): timeouts, retries, isProduction; getEnvName() from TEST_ENV; readEnvList() for lists like PAYMENT_UTILITY_ENVS.

WHY IT MATTERS Unknown environment names fail clearly; lists never accept a production environment.

src/config/portals.ts

PURPOSE Portals (investor, employee) with their login strategy; URLs from <ENV>_<PORTAL>_URL.

WHY IT MATTERS Same app on UAT, but modelled separately so either can move or change login without affecting tests.

src/config/users.ts

PURPOSE Users by key (investor1, employee2 ...); credentials from <ENV>_<KEY>_USERNAME / _PASSWORD.

WHY IT MATTERS Any investor<N> / employee<N> works once its credentials are in .env; the prefix decides the portal.

src/config/run-options.ts

PURPOSE Per-run options: REQUEST_NUMBER, ALLOW_SUBMIT.

WHY IT MATTERS Change what a run does without editing code.

src/config/safety.ts

PURPOSE Production guards: @prod-safe tag, ALLOW_PROD_RUN, assertMutationsAllowed().

WHY IT MATTERS Three independent guards so one mistake is never enough to touch production.

src/config/errors.ts / index.ts

PURPOSE ConfigError (human-readable "what is missing and where") and barrel export.

WHY IT MATTERS Configuration problems fail early with a clear message.

Test data

`src/data/new-exploration-request.data.ts`

PURPOSE Defaults for a NEW request (steps 1–2): service type, category, duration, minerals, geologist, KML regions, size, attempts, map timeout.

WHY IT MATTERS Every value was accepted on TEST. Override with test-data/common|<env>/new-exploration-request.json.

`src/data/exploration-license.data.ts`

PURPOSE Defaults for wizard steps 3–8 (popup values by field label, financial, social, environmental).

WHY IT MATTERS Comments name the rule each value respects. Override with test-data/.../exploration-license.json.

`src/data/test-data.ts`

PURPOSE loadTestData: merges test-data/common/<name>.json with test-data/<env>/<name>.json.

WHY IT MATTERS Environment-specific data without code changes; secrets stay in .env.

Utilities

src/utils/os-waits.ts

PURPOSE OutSystems server-call sync: waitForScreenService, clickAndWaitForServer, trackServerActivity().idle().

WHY IT MATTERS Every data fetch is a POST to …/screenservices/<Module>/<Flow>/<Screen>/<Action>. Waiting for that call is far more reliable than networkidle or sleeps.

Locator	What it finds
`waitForResponse(isScreenService)`	A specific OutSystems server action.

src/utils/field-helper.ts

PURPOSE Field helpers for wizard steps: fieldByLabel, fillIfEmpty, selectIfEmpty, checkRadioByLabel, checkAll, uploadIfEmpty.

WHY IT MATTERS Never overwrite existing values; each helper reports filled vs. kept.

Locator	What it finds
`getByLabel(/^\s*\*?\s*<label>\s*$/)`	Field by its label (optional * for mandatory).
`getByRole('radio', { name }) / label`	Radio by label.
`getByRole('checkbox')`	Checkboxes to tick.
`.dga-uploaded-file / input[type=file]`	Attachment slot: already uploaded / upload input.

src/utils/validation.ts

PURPOSE Reads inline validation messages, splits toast rules, WizardValidationError.

WHY IT MATTERS Failures say exactly which rules the app reported.

Locator	What it finds
`.validation-message, span[class*="validation"] (visible)`	Inline field errors.

src/utils/kml-generator.ts

PURPOSE Generates a KML rectangle (clockwise) inside inland Saudi regions; toDms for degree/minute/second.

WHY IT MATTERS Inland regions only (no sea or neighbours). Clockwise rings are required by the GIS map.

src/utils/data-generator.ts

PURPOSE Valid Saudi national id, mobile number and test e-mail.

WHY IT MATTERS Each generator encodes a format rule enforced by the app.

src/utils/logger.ts / unique.ts

PURPOSE Tagged console logger ([GIS], [Workflow], …) and unique names / suffixes.

WHY IT MATTERS Readable run logs; unique values prevent collisions between parallel runs.

Scripts and project files

`scripts/check-env.ts (npm run env:check)`

PURPOSE Shows which variables are set or missing per environment.

WHY IT MATTERS Never prints usernames or passwords.

`scripts/clear-auth.ts (npm run auth:clear)`

PURPOSE Deletes saved sessions (.auth/).

WHY IT MATTERS Forces fresh logins.

`scripts/generate-kml.ts (npm run kml)`

PURPOSE Generates KML files into test-data/kml/ (region, size, count, seed).

WHY IT MATTERS Manual location tests.

`scripts/record.ts (npm run record)`

PURPOSE Records a scenario like codegen, maximised, cookie banner accepted.

WHY IT MATTERS Fast way to explore a new screen.

`.env / .env.example`

PURPOSE URLs, credentials and options per environment (TEST_ENV, WORKERS, HEADLESS, PAYMENT_UTILITY_ENVS ...).

WHY IT MATTERS .env is git-ignored (secrets); .env.example documents every key.

`.vscode/settings.json, extensions.json`

PURPOSE Playwright Test for VSCode: TEST_ENV for runs from the Testing sidebar, no browser reuse.

WHY IT MATTERS Run / debug tests from the editor.

`package.json, tsconfig.json`

PURPOSE npm scripts (test:test, test:uat, test:new-exploration, report ...) and TypeScript path aliases (@pages, @flows ...).

WHY IT MATTERS Entry points for running; aliases keep imports short.

Appendix A — Server calls the tests wait on

Call	Meaning
`ActionSave_licenseInfo`	Step 1 saved — returns LicenseReqId (new request number).
`ActionBinaryDataToText`	KML file parsed into coordinate rows.
`…/FeatureServer/87|88|89/applyEdits (GIS)`	Location saved after "تأكيد" (3 sequential calls).
`DataActionCheckLocationIsPushed`	App checks the location was saved before leaving step 2.
`DataActionGetLRData`	Investor Requests search.
`DataActionGetData`	Employee Task Management search.
`ActionAssign_Self`	Employee assigns a task to self.
`ActionBill_Req`	Payment notification — Status S000 = paid.

Appendix B — Known traps and how the code handles them

Trap	Handling
GIS "تأكيد" enabled too early	Clicked only after it stays enabled with no error for 4 s.
GIS save is slow	Wait for "نجاح" after "تأكيد" (up to GIS_MAP_TIMEOUT_MS); moving on early → "رجاءاً تحديد الموقع".
Location overlap	"تقاطع المواقع" → new KML in the next region, up to maxAttempts.
Same user logged in twice at once	Login lock: one worker logs in, others reuse the session.
Popup upload inputs disappear while re-rendering	uploadAll counts inputs once and waits for each to be consumed.
Toasts stack	Compare count() before and textsSince() after an action.
UAT employee menu	"إدارة المهام" hidden → Task Management opened by URL.
UAT payment provider id	Page note "UAT: 004" is wrong; template value (002) works.